

Ammar Altanany

Full-Stack Developer | Golang, Node.js, TypeScript, React | AI-Integrated Systems

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PROFESSIONAL SUMMARY

Full-Stack Developer with 2 years of freelance experience and a completed one-year contract as a full-stack developer on Probot, a Discord bot, building scalable, event-driven systems with Golang (Go), Node.js, TypeScript, and React. Specialized in concurrent backend systems in Go (goroutines, channels, worker pools), Discord.js bot ecosystems, and AI-integrated automation platforms built on Redis, MongoDB, and PostgreSQL. Delivered production-grade web applications, REST APIs, authentication systems, and real-time infrastructure for clients, including a custom AI customer-service agent and a high-throughput distributed job-processing engine.

TECHNICAL SKILLS

Languages: Golang (Go), TypeScript, JavaScript (ES6+), HTML5, CSS3

Backend: Golang (Gin, Fiber), Node.js, Express.js, RESTful API Design, Event-Driven Architecture, Microservices, Concurrent and Asynchronous Programming, Goroutines, Channels, JWT/OAuth Authentication, Middleware, WebSockets, Webhooks

Frontend: React.js, Tailwind CSS, Responsive UI Design, State Management, REST and WebSocket Integration

Databases: MongoDB, Mongoose, PostgreSQL, Redis (Streams, Caching, Pub/Sub, Distributed Locking)

DevOps and Monitoring: Docker, Docker Compose, Prometheus, Grafana, Git, GitHub

AI and Automation: LLM Integration, Embeddings, Semantic Search, Prompt Engineering, WhatsApp Business API Automation, AI Customer Service Systems

Discord Ecosystem: Discord.js, DiscordGo, Discord API, Bot Architecture, Rate Limiting, Moderation Systems

Tools and Practices: Postman, npm, Clean Code (DRY/SOLID), Performance Optimization

PROFESSIONAL EXPERIENCE

Full-Stack Developer, Probot (Discord Bot)

Sep 2025 – Sep 2026

Remote | One-year contract, completed at the start of September 2026

- Worked as a full-stack developer on Probot, a Discord bot, for a full one-year contract that concluded at the start of September 2026.
- Built and maintained bot features and backend services across the full stack, working with Discord API events, commands, and data storage.
- Delivered features and fixes to a live Discord bot in production, applying clean-code practices and performance-minded design.

Full-Stack Developer, Freelance (Independent Contractor)

2024 – Present

Remote

- Built concurrent, distributed backend systems in Golang using goroutines and channels for high-throughput, low-latency task processing.
- Engineered full-stack systems with React, Node.js, Express, and Golang for freelance clients, including secure authentication, complex data transactions, and high-performance middleware.
- Designed responsive React frontends paired with RESTful and real-time APIs, delivering complete end-to-end products from UI and system design through deployment and monitoring.
- Integrated AI features into production workloads, including a custom AI customer-service agent and an AI semantic content scanner using LLM prompting and embeddings.
- Built and maintained Discord.js bot ecosystems with Redis queues, MongoDB, and third-party APIs for real-time event routing, moderation, and automation.
- Designed MongoDB and PostgreSQL schemas and Redis key structures to ensure data integrity and low-latency queries.

PROJECTS

GoFlow: Distributed Task Queue and Worker System

Golang, Redis, PostgreSQL, Gin/Fiber, Docker, Prometheus, Grafana

- Built an asynchronous distributed job-processing engine in Golang with concurrent execution using goroutines, channels, and context cancellation.
- Implemented an adaptive worker-pool scaling algorithm driven by real-time queue depth, with exponential-backoff retries and dead-letter queues (DLQ) for failed jobs.
- Added Redis-based distributed locking and deduplication to prevent duplicate job execution across multiple service instances.
- Designed REST management endpoints (Gin/Fiber) and integrated Prometheus and Grafana to monitor queue latency, error rates, and worker utilization.

Ultimate Bot: High-Concurrency Discord Bot

Golang, DiscordGo

- Built a high-concurrency Discord bot in Golang optimized for low latency and reduced CPU and memory usage.
- Used Go's lightweight concurrency model to process webhooks and WebSocket payloads under simulated high load.

DiscordGuard-Core: Discord Security Bot

TypeScript, Node.js, Redis Streams, MongoDB, Discord.js

- Built a backend-only, event-driven Discord security bot that separates event ingestion from processing.
- Developed a Gateway service that writes events to a Redis Stream in under 1 ms, preventing gateway heartbeat drops under heavy load.
- Built worker pools with consumer groups running three protection layers: a Redis rolling-window rate limiter, a sandboxed link resolver using Levenshtein-distance similarity, and an AI semantic scanner using embeddings with a local regex fallback.
- Executed real-time moderation actions (bans and mutes) through the Discord API with permanent logging to MongoDB.

AI Customer Service Automation: WhatsApp AI Agent

Node.js, WhatsApp Business API, LLM, MongoDB

- Built a fully custom-coded AI agent (no low-code tools such as n8n) that responds to customer WhatsApp messages in real time.
- Designed a multi-tenant architecture so the agent can be trained and customized per client company with business-specific knowledge.
- Integrated the WhatsApp Business API for two-way messaging and implemented an LLM pipeline with prompt and context management for accurate, low-latency replies.

Edgy-Ticket: Customer Support Ticket Bot (Freelance)

Node.js, Discord.js, MongoDB

- Built and deployed a high-throughput support ticket bot for a client with dynamic channel creation, real-time transcript generation, secure permission handling, and persistent storage for logs and analytics.

Full-Stack E-Commerce and Management Application

Node.js, Express, React, MongoDB

- Developed a full-stack web application with a secure REST API and a responsive React frontend.
- Implemented JWT authentication and protected routes, with a scalable database design for users, products, and real-time updates.

EDUCATION

Bachelor's Degree (In Progress)

Helwan International Technological University (HITU)

ADDITIONAL INFORMATION

- Strong grasp of concurrent and asynchronous programming, event loops, and server-side logic, with hands-on frontend experience.
- Self-motivated, fast learner, with strong problem-solving and system-planning skills across the full stack.